Industry-Oriented Full Stack E-Commerce Web Application

Objective

To design and develop a full stack e-commerce web application that allows users to browse products, manage a shopping cart, and place orders using MERN/MEAN stack technologies.

Problem Statement

E-commerce platforms must manage product listings, user carts, and order processing efficiently. Poor integration between frontend and backend can lead to slow performance and failed transactions. This project aims to build a scalable and efficient system.

Technologies Used

- Frontend: React.js / Angular
- Backend: Node.js, Express.js
- Database: MongoDB
- API: REST APIs
- Tools: Git, Postman

System Architecture

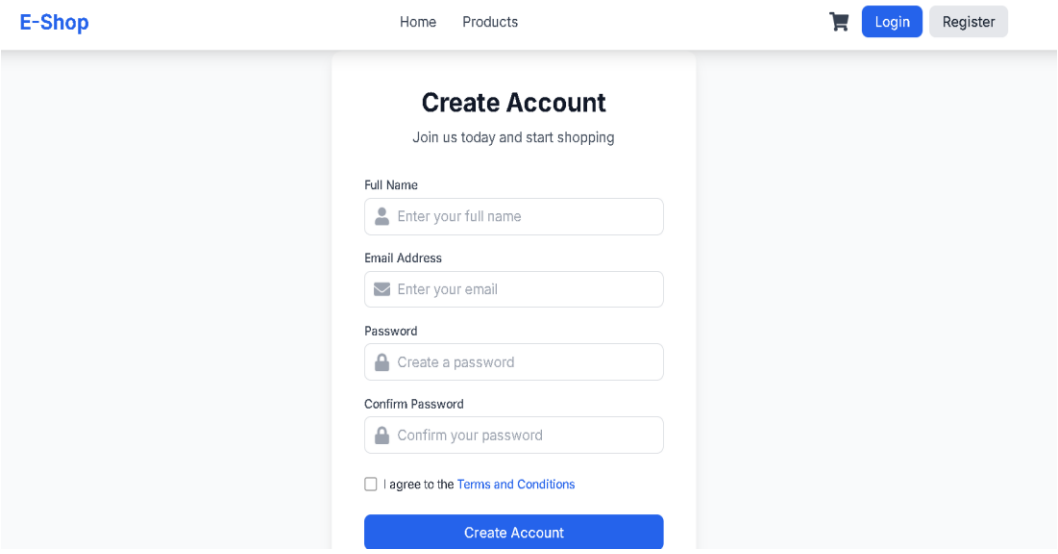
The system follows a **client-server architecture**:

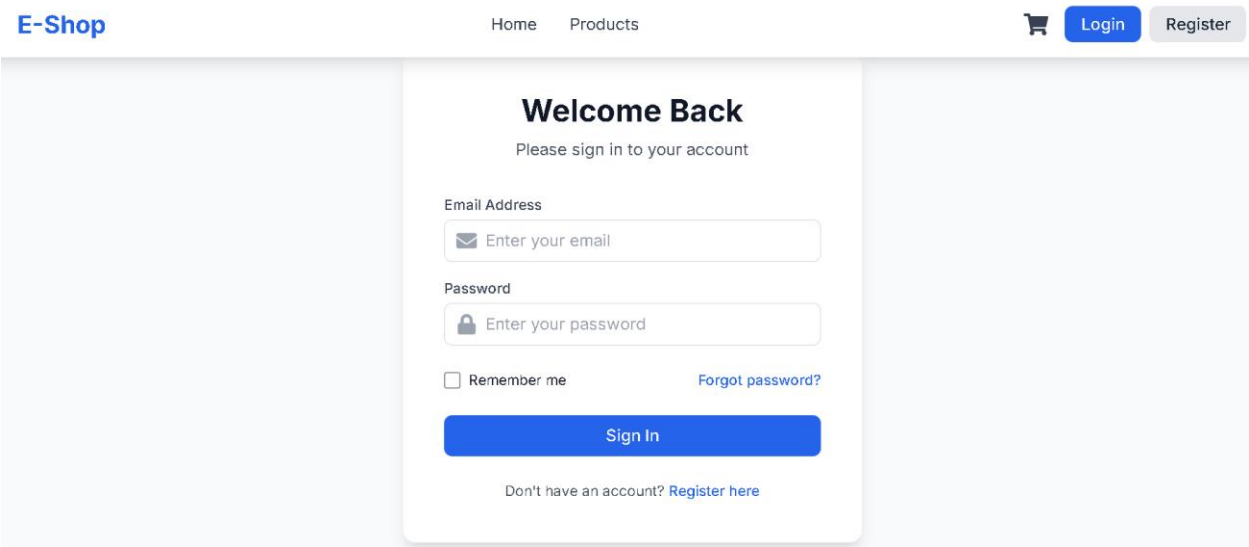
- Frontend sends requests to backend APIs
- Backend processes logic
- Data is stored/retrieved from database
- Response is sent back to frontend

Features Implemented

User Features

- User registration and login
- Product browsing
- Add to cart
- Place order
- View order history





Product Features

- Product listing
- Category management

Order Features

- Order placement
- Order tracking

API Endpoints

- POST /register
- POST /login
- GET /products
- POST /cart
- POST /order

- GET /orders

Database Design

Collections:

Users

- name
- email
- password

Products

- name
- price
- category

Orders

- userId
- products
- totalAmount
- status

Security Features

- Password encryption
- Authentication system
- Protected routes

Testing

Testing was done using Postman to verify:

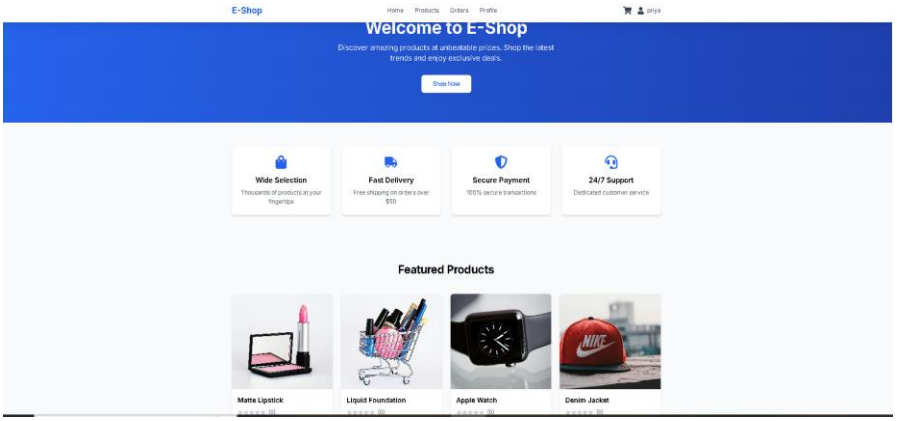
- API responses
- Authentication
- Order processing

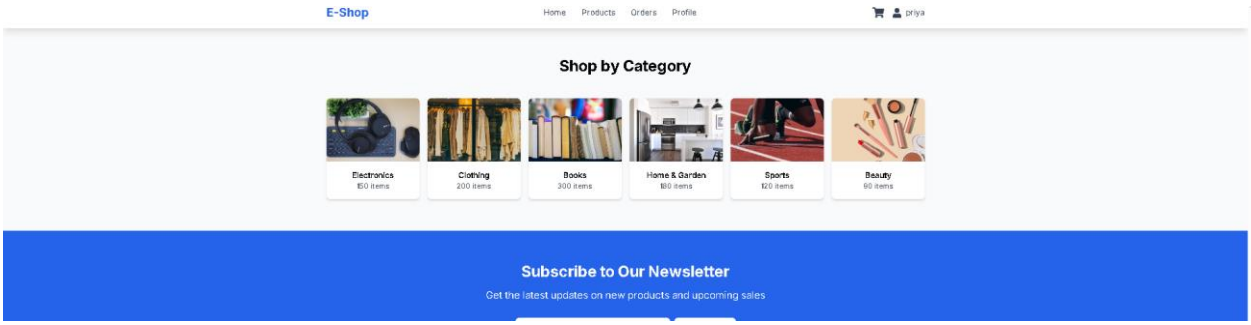
Deployment

- Frontend: Netlify / Vercel
- Backend: Render / Railway

Output

- Home page
- Product page
- Cart page
- Order page




E-Shop
Home
Products
Orders
Profile
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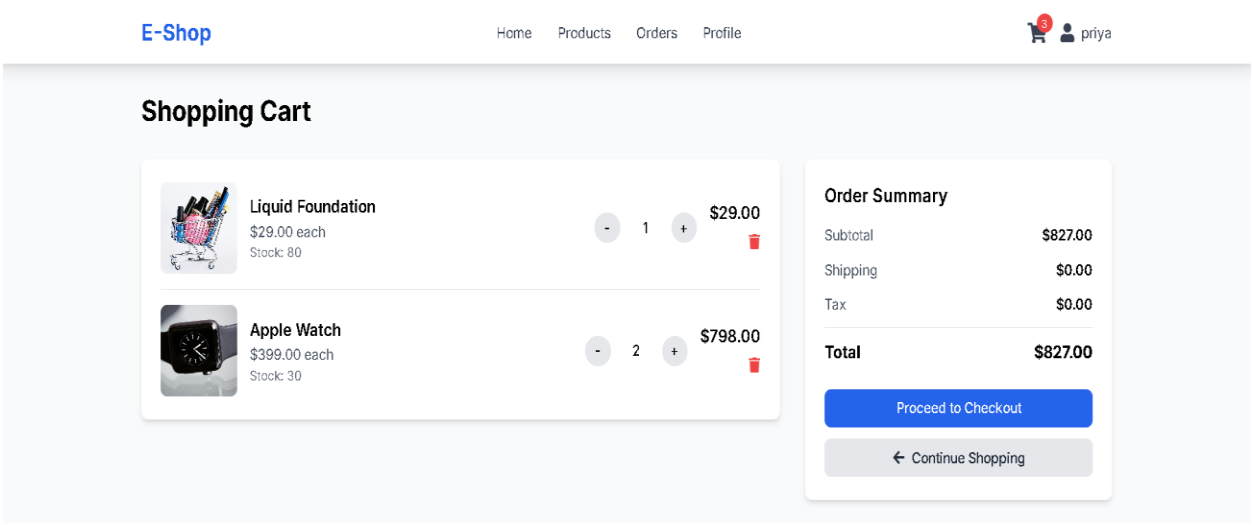
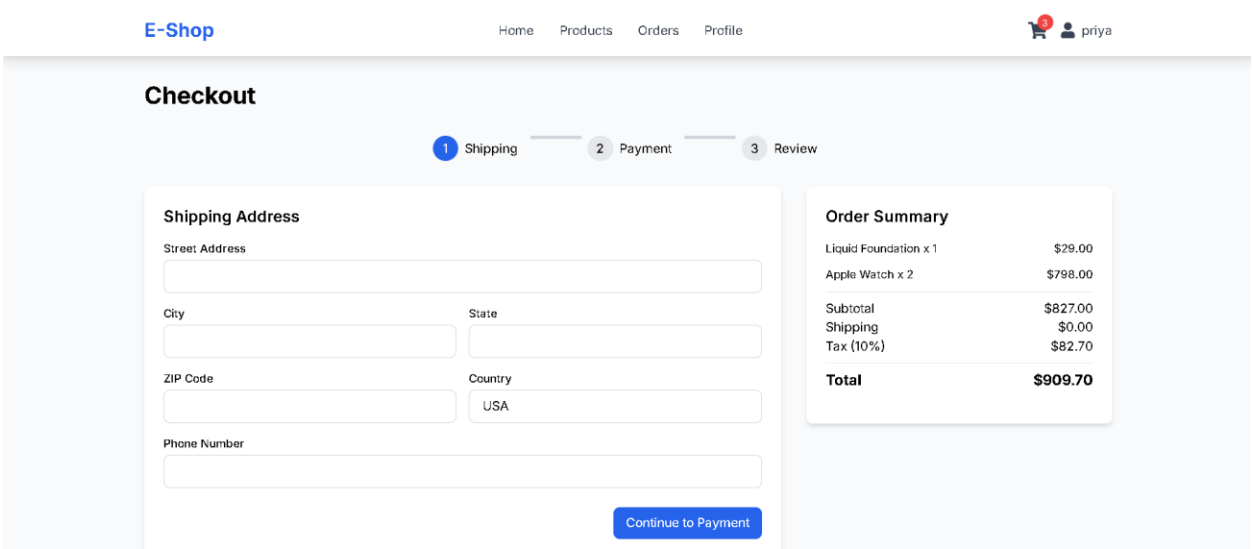
Shop by Category
Electronics
150 items
Clothing
200 items
Books
300 items
Home & Garden
180 items
Sports
120 items
Beauty
90 items
Subscribe to Our Newsletter
Get the latest updates on new products and upcoming sales


E-Shop
Home
Products
Orders
Profile
3
priya
Shopping Cart
Liquid Foundation
$29.00 each
Stock: 80
-
1
+
$29.00
Apple Watch
$399.00 each
Stock: 30
-
2
+
$798.00
Order Summary
Subtotal
$827.00
Shipping
$0.00
Tax
$0.00
Total
$827.00
Proceed to Checkout
Continue Shopping


E-Shop
Home
Products
Orders
Profile
3
priya
Checkout
1 Shipping
2 Payment
3 Review
Shipping Address
Street Address
City
State
ZIP Code
Country
USA
Phone Number
Continue to Payment
Order Summary
Liquid Foundation x 1
$29.00
Apple Watch x 2
$798.00
Subtotal
$827.00
Shipping
$0.00
Tax (10%)
$82.70
Total
$909.70

Conclusion

The project successfully demonstrates a full stack e-commerce system with product management, cart functionality, and order processing. It provides hands-on experience in building real-world applications.